Related Work:

Title	Author	Year	Dataset	Application Area	Method	Attention Type	Metric	Strengths	Limitation	Research Gap	Citations
Multi-class rice seed recognition based on deep space and channel residual network combined with double attention mechanism	Zhang et al.	2025	36 rice seed varieties, 14,040 images (China)	Rice seed variety classification	RSCD-Net (SCR-Block + A^2Net)	Hybrid (Spatial+Channel reconstruction + Double/Bilinear attention)	81.94% acc	Extensive ablation study; feature extraction heatmaps validate effective attention localization.	Overall accuracy (81.94%) remains challenging due to fine-grained 36-class classification, with several difficult classes showing relatively low recognition performance.	No CBAM-style single-block comparison; no cross-dataset validation	[1]
Real-time segmentation and phenotypic analysis of rice seeds using YOLOv11-LA and RiceLCNN	Zhang, Song et al	2025	25,000 rice seeds (9-class) + public datasets	Rice seed detection+classification	YOLOv11-LA + RiceLCNN	SE (channel) + CBAM in detector	89.78%; 96.32% (cross-val)	Beats ViT empirically; edge-deployable	Lower acc, hard fertilizer-treatment task	No SE ablation in classifier	[2]
Enhancing agricultural research with an Attention-Based Hybrid Model for precise classification of rice varieties	Prova	2025	20 Bangladeshi rice varieties, 27,000 images	Rice variety classification	Attention-CNN + CBAM	Hybrid (Channel + Spatial, CBAM)	91.97% (Acc) / 91.92% (F1)	Strong statistical validation (Kappa, MCC, Jaccard)	Classical classifiers show overfitting (100% train acc)	No explainability (Grad-CAM/LIME); single split, no CV	[3]

RiceSeedNet: Rice seed variety identification using deep neural network	Rajalakshmi et al.	2024	13 Tamil Nadu rice seeds (13,000) + public 8-class rice grain (8,000)	Rice seed/grain variety classification	Vision Transformer (ViT, patch-32)	Self-attention (Transformer, no CNN backbone)	97% (own); 99% (cross-val)	Strong cross-dataset generalization; outperforms CNN baselines	No explicit channel/spatial attention module (e.g. CBAM); discards imperfectly segmented seeds; underperforms in darker lighting	No comparison vs. CNN+CBAM/SE hybrids	[4]
Identification of Maize Seed Varieties Using MobileNetV2 with Improved Attention Mechanism CBAM	Ma et al.	2023	11 maize seed varieties, 3,938 images	Seed variety classification (fine-grained)	I_CBAM_MobileNetV2 (parallel CBAM)	Hybrid (Channel + Spatial, parallel-connected CBAM)	98.21% acc	Lightweight (25.1MB, mobile-deployable); Grad-CAM visual proof of improved attention spread	Single crop, controlled lab background only; smaller dataset	Not tested on multiple crops; no comparison with newer transformer-based attention (only CNN-based CBAM/SE variants tested)	[5]

References

[1] T. Zhang, C. Zhang, Z. Yang, M. Wang, F. Zhang, D. Li, and S. Yang, "Multi-class rice seed recognition based on deep space and channel residual network combined with double attention mechanism," PLoS ONE, vol. 20, no. 5, e0322699, May 2025, doi: 10.1371/journal.pone.0322699.

[2] D. Zhang, S. Song, J. Liu, W. Xu, and N. Xiayidan, "Real-time segmentation and phenotypic analysis of rice seeds using YOLOv11-LA and RiceLCNN," Frontiers in Plant Science, vol. 16, 1673143, Dec. 2025, doi: 10.3389/fpls.2025.1673143.

[3] N. N. I. Prova, "Enhancing agricultural research with an Attention-Based Hybrid Model for precise classification of rice varieties," International Journal of Cognitive Computing in Engineering, vol. 6, pp. 412–430, 2025, doi: 10.1016/j.ijcce.2025.02.002.

[4] R. Rajalakshmi, S. Faizal, S. Sivasankaran, and R. Geetha, "RiceSeedNet: Rice seed variety identification using deep neural network," Journal of Agriculture and Food Research, vol. 16, 101062, 2024, doi: 10.1016/j.jafr.2024.101062.

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